Geophysics III

Derivations Tutorial

1 Introduction

What are derivations and why do we do them?
A derivation is the process of developing a relationship between one or more variables that provide physical insight into a scientific problem. These can be quite formal, but are often used to develop a mathematical formulation to predict or model observations. Derivations can also be used to develop corrections to data (you'll see the end result in the gravity corrections).

Why do I make you do derivations?
Some of you will enjoy them, but for the rest, I do have very good reasons for having you perform derivations. I can give you a fish, or teach you how to catch one. If you learn how to do a derivation it will give you the experience that will help you follow along when you read papers involving derivations and perhaps help you gain confidence to do them in the future if the need arise.

The other reason I have you do derivations is that it will help you gain insights into a problem that you otherwise might not get from a single equation. Often a derivation starts from a relatively general point and through a series of approximations and/or constraints arrives at a specific solution. The solution may not be true in all cases as a result of these assumptions and constraints. Seeing, rather than being told, why the solution is limited is an important part of understanding a physical process. I want you to gain the fundamental insight that may be useful in the future, while the equations we derive may not.

2 How to do a derivation

For this course, I want you to be as organized and prescriptive as possible when setting up and then deriving your solutions. Being methodical will help make the process of easier. It also makes derivations easier to read—and easier to assign partial credit! It may not be necessary to do each of these steps every time, but these are the generalized steps and reasons why they are a good idea. There are 4 steps. To remember them you can use the acronym: $\mathbf{D^3C}$, which stands for **d**raw, **d**efine, **d**erive, and **c**heck. **Do NOT input values before you complete the derivation.**

1. **Draw** a picture and annotate the drawing with the symbols you will use in the derivation.

   Drawing a picture can help focus the problem, but also gives you an opportunity to see all the parts and how they relate to one another.

2. **Define** the symbols and give their values and units, including physical constants. You can do this as a table or as a simple list. This is also a good spot to identify the key assumptions and/or constraints that you may know *a priori*. See how I have done the example below. You might leave a few lines space below the list in the event that you need to add variables at a later point.

Making a list of all the symbols and their meaning improves the readability in case you use a different symbol than someone else might for the same variable. It also gives you the opportunity to identify the knowns and unknowns which can help as you decide what you need to do to solve the problem.

It is best to use symbols by convention of the field within which they are used so that it avoids confusion.

3. **Derive** a key result, taking care to keep the equations in symbolic notation (don't plug in values). Enumerate key equations that you know you will need up front. This includes any systems of equations, initial and boundary conditions. Number these equations as it will make it easier to refer to them later. Identify the key steps, locations where: assumptions/constraints are applied; approximations are made; and a shift in focus is made (e.g. you start deriving a secondary result before continuing). While you are working up your derivation, think about where you where you want to go and what you need to do to get there.

   This not only makes it easier to assign partial credit, but also means that the solution you have at the end can be used for any set of inputs, not just the ones you currently have. If we always inserted values at the earliest possible point, we would have to rederive the result every time we wanted to change our inputs.

4. **Check** your answer. Once you have a symbolic solution YOU ARE NOT DONE. Since you may not know what you are looking for, there are several ways to check your solution. First, perform a dimensional analysis (check units) if the units work out properly you've cleared the simplest hurdle. Next, make sure the solution satisfies the initial and boundary conditions. If you know the solution is asymptotic, make sure it satisfies those limits (e.g., if the result should approach zero as position or time approaches infinity). Finally, you can graph the result as a final test, but generally the first two are sufficient.

   Does the solution make sense. Generally, if the solution satisfies the applied constraints and the units work out properly you likely have the correct solution to the problem. If you do the test and it doesn't satisfy the constraints or dimensional analysis, make a note of it and try to identify the location where things most likely went awry. Doing so will vastly improve your partial credit when things don't go as planned.

   Checking your answer is another way to get partial credit if your solution isn't correct. If you can't figure out the answer to a problem, but check the solution and can articulate why your answer cannot possibly be correct you will earn some points you would have otherwise lost.

5. Once you are satisfied with your result, input values if necessary and circle your solution (it helps me know when you think you are done). I also encourage you to discuss your derivations with someone else since it is one of the best ways to make sure that it is clear and complete. Try them yourself, but I encourage you to work in groups to find the solutions.

3 Example

We will be doing similar isostasy (buoyancy related) problems in a couple weeks so don't worry if you don't understand the entire derivation. You will soon.

Problem:
What is the height mantle root necessary for an Airy model of isostatic compensation? Write in terms of the density contrast between the crust and mantle (i.e. $\Delta\rho = \rho_m - \rho_c$).

Answer:
Draw
We begin by drawing a picture of the system and labeling the important components.

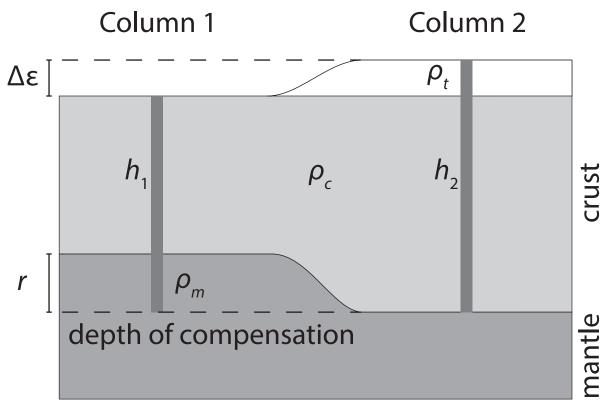

Figure 1: Isostatic columns.

Note that the figure is cartoonish. While there are terms that labeled that will eventually be canceled out, we need them as place holders so they are initially defined.

Describe
Knowns:
elevation, [m]: $\Delta\varepsilon$
upper crustal density, [kg m^{-3}]: ρ_t
density contrast mantle to crust, [kg m^{-3}]: $\Delta\rho = \rho_m - \rho_c$
gravitational acceleration, [m s^{-2}]: g

Unknowns:
height of column 1, [m]: h_1
height of column 2, [m]: h_2
thickness of mantle root, [m]: r

Solve for r, [m].

Derive
We start by equating columns pressures at a common depth of compensation,

$$\underbrace{\rho_c g h_1 + \rho_m g r}_{\text{column 1}} = \underbrace{\rho_t g \Delta\varepsilon + \rho_c g (h_2 - \Delta\varepsilon)}_{\text{column 2}}. \tag{1}$$

Because g is in every term in Equation 1, it can be canceled out,

$$\rho_c h_1 + \rho_m r = \rho_t \Delta\varepsilon + \rho_c (h_2 - \Delta\varepsilon). \tag{2}$$

Assumption: The density of air is negligible so we do not need a pressure term to account for the additional pressure of the air column. To prove that we don't need an air term, we can do a back of the envelope calculation to compare the relative pressures of a lithospheric column to that of the air column. The density of air is on order 2 kg m^{-3} and the density of rock is on the order of 3000 kg m^{-3}. The height of the air column is on the order of 1 km and the height of the crust on the order of 30 km. Hence, the ratio of the two terms is

$$\begin{aligned}\frac{\rho_c h_c}{\rho_{\mathrm{air}}\Delta\varepsilon} &= \frac{(3000\ \mathrm{kg\,m^{-3}})(30\ \mathrm{km})}{(2\ \mathrm{kg\,m^{-3}})(1\ \mathrm{km})} \\ &= \frac{90000}{2} \\ &= 4.5\times 10^4.\end{aligned}$$

It really doesn't matter if the heights vary considerably, the end result is similar, the pressure due to a column of crust is on the order of 10,000 times larger, so we can effectively ignore the influence of air above the column.

Next, we equate column heights

$$h_2 = \Delta\varepsilon + h_1 + r. \tag{3}$$

Next, we can substitute Equation 3 into the modified pressure Equation 2. And after some algebra, we arrive at a solution for the root,

$$\begin{aligned}\rho_c h_1 + \rho_m r &= \rho_t\Delta\varepsilon + \rho_c(\Delta\varepsilon + h_1 + r - \Delta\varepsilon), \\ \rho_m r - \rho_c r &= \rho_t\Delta\varepsilon + \rho_c h_1 - \rho_c h_1, \\ (\rho_m - \rho_c) r &= \rho_t\Delta\varepsilon, \\ r &= \frac{\rho_t}{\rho_m - \rho_c}\Delta\varepsilon.\end{aligned}$$

Recall from the problem statement that we are looking for a solution in terms of $\Delta\rho$,

$$\boxed{r = \frac{\rho_t}{\Delta\rho}\Delta\varepsilon}.$$

Check

Let us check the units,

$$\begin{aligned}[\mathrm{m}] &\stackrel{?}{=} \frac{[\mathrm{kg\,m^{-3}}]}{[\mathrm{kg\,m^{-3}}]}[\mathrm{m}], \\ [\mathrm{m}] &= [\mathrm{m}].\end{aligned}$$

Our units check out. Unfortunately, for this derivation there aren't any additional tests we can use to verify the validity of the solution.